THE SEPARABLE ELEMENTS OF AN EXTENSION FORM A SUBEXTENSION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file fields.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: The separable twin of the purely inseparable subextension lemma (b7 card L48372) and the algebraic subextension lemma (b7 card L48374).

Editorial note. The label lemma-separable-elements names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Chapter lemma-separable-goes-up (delivered card L48370): separability goes up in towers, so beta separable over k is separable over k(alpha)
- Chapter lemma-separably-generated-separable (source lemma): an extension finitely generated by separable elements is a separable extension
- Definition in this chapter: an extension is separable when every one of its elements is separable over the base

Source reference. Original reference 09H2: sources/stacks-fields/pinned-original-source.tex, source lines 1404–1404. Prerequisite pointer invoked at 1659; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 030X: sources/stacks-fields/pinned-original-source.tex, source lines 1560–1560. Prerequisite pointer invoked at 1660; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let E/k be a field extension. Then the elements of E separable over k form a subextension of E/k.*

Proof. Let $\alpha, \beta \in E$ be separable over k. Then β is separable over $k(\alpha)$ by Lemma 09H2. By Lemma 030X (applied with $n = 2$, $\alpha_1 = \alpha$, and $\alpha_2 = \beta$) we see that $k(\alpha, \beta)$ is separable over k. □

Date: stacks-project:fields.tex:local-snapshot-20261012; GNU FDL 1.2 or later.